MIN KHANT ZAW

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EDUCATION

Massachusetts Institute of Technology (MIT) | Cambridge, MA Expected May 2028

- Bachelor of Science/Master of Engineering in Electrical Engineering & Computer Science, GPA: 4.8/5.0
- Relevant Coursework: Solid-State Circuits, Computation Structures, Digital Systems Laboratory

SKILLS

Electronics: PCB Design, Analog Circuit Design, IC Layout, Altium Designer, LTspice, Cadence Virtuoso, Arduino, FPGA
Programming: Python, C, Assembly, Verilog, Minispec (HDL), MATLAB, Robot Operating System (ROS), Git

WORK EXPERIENCE

Rheyo, Cambridge, MA | Electrical Engineering and Prototyping Intern June 2026 – Aug 2026

- Designed the custom PCB (schematic capture, layout, component selection) for a proprietary electrically assisted teeth-whitening platform connecting an ESP32-P4 module, MIPI DSI display, backlight driver, and PIC host interface
- Ported a Python-based controller on Raspberry Pi to C on ESP32-P4, rendering a state-driven UI on a 720x1280 MIPI DSI panel from UART messages sent by a PIC host controller
- Reduced BOM cost by 88% and boot time from 30s to <2s, cut current draw by 98% (0.8 A to 14 mA) for ~57× longer estimated battery life, and simplified assembly by migrating from a Raspberry Pi to a custom ESP32-P4 design

Radius Lab, Cambridge, MA | Undergraduate Researcher Sept 2025 – Present

- Research state-of-the-art ground-penetrating radar technologies to develop a PCB-based search-and-rescue system
- Simulate a published frequency-modulated continuous-wave (FMCW) radar design in Keysight ADS and design a PCB with Altium Designer to analyze signal behavior and system performance

PROJECT EXPERIENCE

Heat Monitor, Cambridge, MA Feb 2026 – May 2026

- Developed wearable heat-monitoring devices that generate a crowdsourced heat map of the MIT campus as users carry the devices on their backpacks, collaborated within a 9-member interdisciplinary team
- Led hardware development of an ESP32-based PCB, including component selection, schematic capture, layout, power management, and testing; achieved <$20 COGS, ±3°F temperature accuracy, and 61-hour battery life per charge

Operational Amplifier Design, Cambridge, MA Nov 2025 – Dec 2025

- Designed a two-stage, Miller-compensated CMOS operational amplifier in Cadence Virtuoso featuring a PMOS differential input pair with cascodes followed by a common source amplifier at the output stage
- Achieved 75.5 dB gain, 200 MHz unity-gain bandwidth, 61° phase margin, 76 dB Common Mode Rejection Ratio (CMRR) at 10 MHz, 0.4 mW power consumption, 309 mV output swing, and 4.7 ns settling time

RISC-V Processor, Cambridge, MA Oct 2025 – Dec 2025

- Designed and implemented a 4-stage pipelined RISC-V processor using Minispec Hardware Description Language (HDL), featuring instruction and data caches, data-bypassing, branch speculation, and hazard stalling mechanisms
- Optimized performance, achieving 274,000 ns runtime on an MNIST-dataset neural-network inference workload

BJT Wideband Amplifier, Cambridge, MA | Project Link Oct 2025

- Designed a wideband amplifier using 2N3904 BJTs through hand analysis, implemented and validated the circuit
- Achieved 40.40 dB gain, 80 mW power consumption, 4.8 V output swing, 44.5 kHz – 4.5 MHz bandwidth using a cascode topology with input and output buffers

LEADERSHIP

Gordon-MIT Engineering Leadership (GEL) Program, Cambridge, MA | Program Website Sept 2025 – Present

- Collaborate with a 7-member team to develop engineering leadership skills such as resilience, resourcefulness, ethics, advocacy, and negotiation by alternating between leadership and team roles in weekly role-based activities